SOP-BAS-ACC-73-HUAWEI-Access-OTN-Failure

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4315

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve a major optical transport failure on the Huawei Access equipment at the Basavanagudi POP. The failure domain in question is attributed to the degradation of SFP/XFP modules or fiber cuts resulting in Rx/Tx power loss. This scenario is specific to the Access role of the Huawei equipment, which relies on the OTN (Optical Transport Network) protocol to establish and maintain optical connections between nodes. The underlying protocol mechanics of OTN involve the use of SONET/SDH (Synchronous Optical Networking/Synchronous Digital Hierarchy) framing to multiplex multiple low-speed signals onto a single high-speed optical interface. In the event of a failure, the OTN protocol will trigger alarms and initiate recovery procedures to restore the optical connection. However, in cases where the failure is attributed to hardware degradation or fiber cuts, manual intervention is required to diagnose and repair the issue.

Pre-requisite Checks

- Task: Verify the site location and access permissions, Description: Determine that the Basavanagudi POP is accessible and that the necessary permissions have been granted to perform the troubleshooting and repair.

- Task: Confirm the redundant link status, Description: Verify that the redundant link is up and functioning correctly to ensure that traffic is being redirected and that the impact of the failure is minimized.

- Task: Gather equipment and documentation, Description: Collect all necessary equipment and documentation, including the Huawei equipment manual, OTN protocol documentation, and any relevant maintenance records.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP module	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display otn link-info, Expected Output: Link information for OTN interface OTU4 0/0/1/1, Description: This command will display the link information for the OTN interface, including the link status, signal quality, and optical power levels.
```

```
Command: display otn alarm-info, Expected Output: Alarm information for OTN interface OTU4 0/0/1/1, Description: This command will display the alarm information for the OTN interface, including any active alarms and their corresponding severity levels.

Command: display sfp info, Expected Output: SFP information for interface 0/0/1/1, Description: This command will display the SFP information for the interface, including the SFP type, status, and optical power levels.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP module, Command: replace sfp 0/0/1/1 with , Description: Replace the degraded SFP/XFP module with a new one to restore the optical connection.

Step 2: Step: Repair the fiber cut, Command: repair fiber cut on interface 0/0/1/1, Description: Repair the fiber cut on the interface to restore the optical connection.

Step 3: Step: Adjust the Rx/Tx power levels, Command: set otn rx-tx-power 0/0/1/1, Description: Adjust the Rx/Tx power levels on the interface to optimize the optical connection.

Post-Check Verification

```
Command: display otn link-info, Expected Output: Link information for OTN interface OTU4 0/0/1/1 (up), Description: Verify that the link is up and functioning correctly after the repair.

Command: display otn alarm-info, Expected Output: Alarm information for OTN interface OTU4 0/0/1/1 (no active alarms), Description: Verify that there are no active alarms on the interface after the repair.

Command: display sfp info, Expected Output: SFP information for interface 0/0/1/1 (normal), Description: Verify that the SFP information is normal and the interface is functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace sfp 0/0/1/1 with , Description: Restore the original SFP/XFP module if the repair failed.

- Step: Restore the original fiber configuration, Command: restore fiber configuration on interface 0/0/1/1, Description: Restore the original fiber configuration if the repair failed.

- Step: Restore the original Rx/Tx power levels, Command: set otn rx-tx-power 0/0/1/1 to original value, Description: Restore the original Rx/Tx power levels if the repair failed.